MILITARY SPECIFICATION

TROUSERS, COLD WEATHER

This specification is approved for use by the Navy Clothing and Textile Research Facility, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for a lined cold weather trouser.

1.2 Classification. The trousers shall be of the following types, classes and sizes (see 6.2):

1.2.1 Type I - Men's

Sizes

Small (27-30) Large (35-38)
Medium (31-34) X-Large (39-42) XX-Large (43-46)

1.2.2 Type II - Women's

Sizes

Small (22-25) Medium (26-29) Large(30-33)

1.2.3 Classes.

Class 1 - Blue 3375, Flame Retardant Plain Weave
Class 2 - Green OG-107, Cotton/Nylon Sateen

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Officier in Charge, Navy Clothing and Textile Research Facility, 21 Strathmore Road, Natick, MA 01760 by using the self addressed standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8415

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## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified.

SPECIFICATIONS

FEDERAL

| | |
|---|---|
| V-B-871 | - Button, Sewing Hole, and Button, Staple (Plastic) |
| V-F-106 | - Fasteners, Slide, Interlocking |
| V-T-280 | - Thread, Cotton Gimp, Buttonhole |
| NN-P-71 | - Pallet, Material Handling, Wood, Stringer Construction, 2 Way and 4 Way (Partial) |
| DDD-L-20 | - Label; For Clothing, Equipage, and Tentage (General Use) |

MILITARY

| | |
|---|---|
| MIL-B-371 | - Braid, Textile (Cotton, Tubular) |
| MIL-E-480 | - Enamel, Baking, Phenol or Urea-Formaldehyde |
| MIL-W-530 | - Webbing and Tape, Textile, Cotton, General Purpose, Natural or in Colors |
| MIL-B-543 | - Buckles, Tongueless and Web Strap |
| MIL-T-3530 | - Thread and Twine, Mildew Resistant, Water Repellent Treated |
| MIL-P-15011 | - Pallet, Material Handling, Wood, Post Construction, 4 Way Entry |
| MIL-C-17155 | - Cloth, Knitted, Nylon Fleece |
| MIL-B-17757 | - Boxes, Shipping, Fiberboard (Modular Sizes) |
| MIL-C-24931 | - Cloth, Plain Weave, Aramid, Water Repellent |
| MIL-C-24932 | - Cloth, Knitted, Flame Resistant, Rayon/Wool Pile |
| MIL-C-43191 | - Cloth, Wind Resistant, Sateen, Cotton and Nylon |
| MIL-T-43548 | - Thread, Polyester Core: Cotton-, Rayon-, or Polyester-Covered |
| MIL-T-83193 | - Thread, Aramid, Spun Staple |

STANDARDS

FEDERAL

| | |
|---|---|
| FED-STD-751 | - Stitches, Seams, and Stitching |

MILITARY

| | |
|---|---|
| MIL-STD-129 | - Marking for Shipment and Storage |
| MIL-STD-147 | - Palletized Unit Loads |
| MIL-STD-1488 | - Provisions for Evaluating Quality of Trousers |



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(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

LAWS AND REGULATIONS

U.S. POSTAL SERVICE MANUAL

(Copies of the manual may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Applications for copies should be addressed to the American Trucking Association, ATTN: Traffic Department, 1616 P Street, N.W., Washington, DC 20036).

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606).

THE COLOR ASSOCIATION OF THE UNITED STATES, INC.

Standard Shades for Buttons - 1966

Standard Shade for Thread - 1968

(Application for copies of color cards should be addressed to the Color Association of the U.S. Inc., 343 Lexington Avenue, New York, NY 10016).



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2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Guide sample. Samples, when furnished, are solely for guidance and information to the contractor (see 6.3). Variation from this specification may appear in the sample, in which case this shall govern.

3.2 First article. When specified, the contractor shall furnish sample units(s) for first article inspection and approval (see 4.3 and 6.2).

3.3 Materials.

3.3.1 Basic material.

3.3.1.1 Class 1. The basic material for the class 1 trousers shall be a aramid plain weave, water repellent fabric conforming to MIL-C-24931. The color shall be shade Blue 3375.

3.3.1.2 Class 2. The basic material for the class 2 trousers shall be cotton/nylon sateen with a florocarbon type water repellent treatment, conforming to class 2 of MIL-C-43191, except that the requirements for infrared reflectance shall not apply. The color shall be shade green OG-107, piece number 3125.

3.3.2 Binding. The material for binding the waist edge, front edge of fly extension and leg bottoms shall be made from the basic material specified in 3.3.1, bias-cut in strips of 1 1/4 (+ 1/8) inches wide. The ends of the strips shall be joined by seam type SSa-1 and stitch type 301 with 10 to 12 stitches per inch. The seams shall finish 1/8 to 3/16 inch wide and shall be spread open.

3.3.3 Lining.

3.3.3.1 Class 1. The material for lining the class 1 trousers shall be flame resistant rayon/wool knitted pile fabric conforming to MIL-C-24932. The color shall be shade blue 3375.

3.3.3.2 Class 2. The lining material for the class 2 trousers shall be knitted nylon fleece conforming to type I of MIL-C-17155.

3.3.4 Thread.

3.3.4.1 Class 1. The thread for the class 1 trousers shall be an aramid thread, conforming to MIL-T-83193. As an alternate, a para-aramid,spun staple thread, size 35 Tex may be used. The color of the thread shall approximate shade Blue, A.R., G.A. 66058. The thread shall meet commerically acceptable colorfastness requirements for aramid thread.



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3.3.4.2 Class 2. The thread for the class 2 trousers for seaming, stitching, bartacks, and buttonholes shall be covered polyester core thread conforming to ticket no. 30, 50 and 70, 2 or 3 ply of MIL-T-43548. The thread for button sewing shall be ticket no. 24, 4 ply of MIL-T-43548. The thread shall be water repellent treated to conform with Type II, class 3 of MIL-T-3530. The color shall conform to shade Olive Drab S-1, C.A. 66022 of the Standard Shades for Thread. The polyester wrapped, polyester core thread is not allowed.

3.3.4.3 Colorfastness. The dyed thread shall show colorfastness to laundering equal to or better than the standard sample when tested as specified in MIL-T-43548. When no standard sample is available, the dyed thread shall show a minimum of "good" colorfastness to laundering when tested as specified in MIL-T-43548.

3.3.5 Gimp. The gimp for reinforcing the buttonholes shall be thread, cotton gimp, buttonhole, conforming to type I or II, size No. 8 of V-T-280. The shade of the gimp shall be as stated in 3.3.4.

3.3.5.1 Colorfastness. The dyed gimp shall show colorfastness to laundering equal to or better than the standard sample when tested as specified in V-T-280. When no standard sample is available, the dyed gimp shall show a minimum of "Good" colorfastness to laundering when tested as specified in V-T-280.

3.3.6 Button, plastic. The button for the front and hip pocket flap closure shall be 30-line, conforming to type II, class D, style 20 and 21 with stud of V-B-871. The button shall have a glossy finish and the color shall match the basic material, as listed in the Standard Shades for Buttons.

3.3.7 Slide fastener. The slide fastener for fly closure shall be interlocking, conforming to type I, style 3, size M, long tab pull, brass chain of V-F-106. All metal components of the slide fastener shall be from the same supplier (see 4.4.1).

3.3.7.1 Tape for slide fastener. The tape for the slide fastener for the class 1 trousers shall be high-temperature resistant aramid and for the class 2 trousers shall be cotton or synthetic or a combination. The tape shall approximate the shade of the basic material and show good fastness to laundering as specified in V-F-106.



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3.3.7.2 Length of slide fasteners. The length of the slide fasteners shall be as follows:

| Sizes - Inches (+ 1/4 inch) | |
|---|---|
| Type I | |
| Small (27-30) | 8 |
| Medium (31-34) - Large (35-38) | 9 |
| X-Large (39-42) - XX-Large (43-46) | 10 |
| Type II | |
| Small (22-25) | 7 1/2 |
| Medium (26-29) | 8 |
| Large (30-33) | 8 1/2 |

3.3.8 Buckle. The buckle for the leg adjustment (take-up) straps shall be type II, style 4, class 1, construction a or b, size 3/4 inch conforming to MIL-B-543, except that the buckle shall have baked-on enamel finish conforming To type I or type II of MIL-E-480 (see 4.4.1). The color shall approximate the shade of the basic material.

3.3.9 Webbing. The material for the leg adjustment (take-up) straps shall be cotton, conforming to type IIa, class 3, 3/4 inch wide of MIL-W-530. The color shall be match the basic material.

3.3.10 Suspender loops (braid). The braid for the suspender loops shall be cotton, tubular, conforming to type V, class 1 of MIL-B-371, except that the colorfastness requirements specified therein shall not apply. The color shall match the basic material.

3.3.11 Label. Each pair of trousers shall have a combination identification/instruction label and a size label conforming to the requirements of 3.3.11.1 and 3.3.11.2. The printing shall show fastness to laundering.

3.3.11.1 Combination label. The combination label shall conform to Type VI, class 15 of DDD-L-20, and shall bear the following inscription and format for the class 1 trousers:

TROUSERS, COLD WEATHER, FLAME RESISTANT, CLASS 1(Example)
FIBER CONTENT: SHELL 100% ARAMID
LINING 65% RAYON/35%WOOL PILE
100% ARAMID BACK
CONTRACT NO: DLA-100-00-0-0000 (EXAMPLE)
NAME OF CONTRACTOR:

LAUNDERING INSTRUCTIONS

1. Wash trousers in hot water ($120^{o}$ - $140^{o}$ F) using detergent.
2. Rinse thoroughly 2 or 3 times in hot water. Tumble dry.
3. DO NOT USE STARCH OR BLEACH. DO NOT DRY CLEAN.



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OR

Follow NAVEDTRA Manual
414-01-45-81
Formula II

The following inscription and format for the class 2 trousers:

TROUSERS, COLD WEATHER, PERMABLE CLASS 2 (EXAMPLE)
FIBER CONTENT: SHELL 56% COTTON/50% NYLON
LINING 100% NYLON
CONTRACT NO: DLA-100-00-0-0000 (EXAMPLE)
NAME OF CONTRACTOR:

The laundering instruction shall be the same as the class 1 trousers.

NOTE: The braid side loops are provided for use with Suspenders, Trousers, M-1950 Stock No. 8440-00-221-0852.

3.3.10.2 Size label. The size label shall conform to Type VI, class 2 of DDD-L-20, and shall bear the following inscription:

SIZE: MEDIUM (31-34) (EXAMPLE)
STOCK NUMBER: 8415-00-000-0000 (EXAMPLE)

3.4 Design. The class 1 trousers shall be made of a water repellent, flame resistant fabric lined with a knitted flame resistant rayon/wool pile fabric. The Type I and II trousers shall be bound at the waist, front edge of fly extension and at edges of each leg opening. The trouser shall have two front quarter top patch pockets and one left back hip patch pocket with a buttoned flap; a slide fastener and buttoned fly closure; six belt loops; and a high back riser horizontally quilted. Each leg shall have an adjustment (take-up) strap assembly located across the outseam near the bottom of the trouser. The class 2 trousers shall be of the same design except that the material shall be a water repellent, wind resistant cotton/nylon sateen cloth and lined with a nylon knitted fleece. (see figures 1 and 2).

3.4.1 Figures. The figures are furnished for information purposes only. In the event of any inconsistencies between the specification and the figures, the specification shall govern.

3.5 Patterns. Standard patterns to be used to cut working patterns will be furnished by the Government (see 6.3). The patterns allow 1/2 inch for all double lapped and double needled seams, except as otherwise indicated in Table I. The working patterns shall be identical to the Government patterns. Neither the Government patterns nor the working patterns shall be altered in any way.



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3.5.1 List of pattern parts. The component parts shall be cut from material as specified and in accordance with the following list of pattern parts:

| Material | Nomenclature of pattern parts | Cut parts |
|---|---|---|
| | **Type I** | |
| Cloth, basic material | Front | 2 |
| | Back | 2 |
| | Fly, Facing (left side) | 1 |
| | Fly, Extension (right side) | 2 |
| | Patch Pocket Front | 2 |
| | Patch Pocket Front, Facing | 2 |
| | Patch Pocket Hip (left side) | 1 |
| | Patch Pocket Flap, Hip | 1 1/ |
| | Bottom Leg Lining Extension | 2 |
| | Back Waist Interlining | 1 |
| Cloth, lining material | Front lining | 2 |
| | Back lining | 2 |
| Template | Belt Loops marker | - |
| | Leg Adjustment (take-up) Strap Assembly marker | - |
| | Hip Pocket | - |
| | **Type II** | |
| Cloth, basic material | Front | 2 |
| | Back | 2 |
| | Fly Facing (right side) | 1 |
| | Fly Extension (left side) | 2 |
| | Patch Pocket Front | 2 |
| | Patch Pocket Front, Facing | 2 |
| | Patch Pocket Hip (left side) | 1 |
| | Patch Pocket Flap, Hip | 1 1/ |
| | Bottom Leg Lining Extension | 2 |
| | Back Waist Interlining | 1 |
| Cloth, lining material | Front Lining | 2 |
| | Back Lining | 2 |
| Template | Belt Loops marker | - |
| | Leg Adjustment (take-up) Strap Assembly marker | - |
| | Hip Pocket | - |

1/ When alternate method of pocket flap construction is chosen, two piece pocket flap shall be cut with seam allowances added (see operation 6b).



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3.6 Stitches, seams and stitching. Stitch, seam and stitching types specified in table I shall conform to FED-STD-751. Whenever two or more methods of seams or stitches are given for the same operation, any one of them may be used. Where stitch type 401 is used, the looper (underthread) shall be on the inside of the trousers. Double needle stitching shall be 1/4 to 5/16 inch gauge with the outer row of stitching 1/16 to 1/8 inch from folded edge. Unless otherwise specified, topstitching and edgestitching shall be 1/16 to 1/8 inch from folded edge. Ends of all seams produced by stitch type 401 shall be caught in other seams or stitchings.

3.6.1 Stitches per inch. The minimum and maximum number of stitches per inch shall be as specified in Table I.

3.6.2 Thread breaks and ends of seams. Ends of all seams and stitchings when not caught in other seams or stitching shall be backtacked not less than 3/8 inch. The ends of a continuous line of stitching shall be overlapped not less than 1/2 inch. Thread breaks (all stitch types) shall be secured by stitching back of break not less than 1/2 inch. Skipped stitches or thread breaks may be repaired by using 301 stitch type.

3.7 Manufacturing operations requirements. The trousers shall be manufactured in accordance with all operations as shown in Table I. The contractor is not required to follow the exact sequence of operations as listed provided the finished trousers are identical to that produced by following the exact sequence as shown in Table I. Unless otherwise indicated, all operations apply to both Type I and Type II trousers.

3.7.3 Abbreviations in table of operations. The abbeviations used in table I are as follows:

| | | |
|---|---|---|
| Stch | - | Stitch |
| in | - | inch |
| Ndl | - | Needle |
| Bob | - | Bobbin |
| Lpr | - | Looper |
| Mchne | - | Machine |
| Brtck | - | Bartack |
| Comrcl | - | Commercial |
| smlr | - | similar |
| Btnhl | - | Buttonhole |
| incl | - | including |
| dbl | - | double |



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TABLE I - CONSTRUCTION OF TROUSERS

| No. | Operation | Stch Type | Seam/ Stch type | Stch in | Thread Ndl | Thread Bob/ Lpr |
|---|---|---|---|---|---|---|
| 1. | Cutting. | | | | | |
| | a. The trousers shall be cut in strict accordance with patterns furnished, which shall show size, placement of notches, perforations, directional lines for cutting, and instructions for the proper assembly of all component parts. | | | | | |
| | b. Directional lines on the fronts and backs may vary from the warp direction by not more than 2 1/2 inches. Measurements shall be taken at top and bottom edges of the pattern from the directional line to the selvage edge (on the even side of lay) of the fabric and the difference between the two measurements shall not exceed 2 1/2 inches. | | | | | |
| | c. The pile or fleece side of the knitted shall be the face side of the lining material. | | | | | |
| | d. Cut all component parts of the basic material from one piece of material, except for the patch pocket front facing, fly facing, fly extension, back waist interlining, binding, belt loops and bottom leg lining extension pieces which may be cut from ends and when so cut shall approximate the shade of the main assembly. | | | | | |
| | e. Cut basic material stripping as follows:<br>- for binding (see 3.3.1.1)<br>- for knee stay 1/2 (± 1/8) inch wide by 2 1/4 (± 1/2) inches long<br>- for belt loops (see operation 15). | | | | | |
| | f. Cut all component parts of the lining from one piece of material. | | | | | |



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TABLE I - CONSTRUCTION OF TROUSERS

| No. | Operation | Stch Type | Seam/ Stch type | Stch in | Thread Ndl | Thread Bob/ Lpr |
|---|---|---|---|---|---|---|
| 2. | Replacement of defective components. | | | | | |
| | a. During the spreading, cutting, and manufacturing process, components having material defects or damages that are classified as defects in MIL-STD-1488 and section 4 shall be removed from production and replaced with non-defective and properly matched components. | | | | | |
| 3. | Shade marking. | | | | | |
| | a. All parts shall be marked, ticketed, or bundled to insure uniform shade, size, and proper assembly throughout the trousers, except those parts cut from ends. Parts cut from ends need not be shade marked, however, they must match the main assembly. | | | | | |
| | b. The use of corrosive metal fastening devices or sew-on tickets is prohibited. | | | | | |
| | c. Adhesive type shade tickets which discolor or adhere to the material upon removal of tickets shall be prohibited. | | | | | |
| | d. The use of ink pad numbering machine, rubber stamp or pencil is acceptable, provided the numbers do not show on the outside of the trousers and whenever possible, numbers shall be covered by the seam allowance. | | | | | |
| 4. | Make front patch pockets. | | | | | |
| | a. Finished appearance. The edge of the pocket shall finish smooth and flat. The openings shall be aligned uniform in length, and shall measure not less than 6 1/4 inches nor more than 6 3/4 inches between tacks for type I, or not less than 6 inches nor more than 6 1/2 inches for type II. The facing shall not be exposed beyond edge of pocket opening. Pockets may be precreased prior to setting. | | | | | |
| | b. Stitch, 3/8 inch from edge, face to face, the facing to the pocket piece. | 301 or 401 | SSe-2(a) | 10-14 | 30<br>30 | 30<br>50 |
| | c. Turn facing to finished position and edgestitch, 3/16 to 1/4 inch from folded edge. | 301 | SSe-2(b) | 10-14 | 30 | 30 |



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TABLE I - CONSTRUCTION OF TROUSERS

| No. | Operation | Stch Type | Seam/ Stch type | Stch in | Thread Ndl | Thread Bob/ Lpr |
|---|---|---|---|---|---|---|
| | d. Fold under raw edge of facing 3/8 inch and topstitch the back and bottom raw edge of facing through pocket piece, 1/16 inch from folded edge. | 301 | LSd-1 | 10-14 | 30 | 30 |
| 5. | Stitch front patch pockets. | | | | | |
| | a. Position pocket piece on the outside of each front as indicated by notches on pattern. Stitch, 1/8 inch from edge, the top and side edges of the pocket pieces to the waist and outseam edges. Stitching on the outseam shall terminate approximately 2 inches above bottom edge of pocket to facilitate creasing. | 301 or 401 | SSa-1 | 10-14 | 30<br>30 | 30<br>50 |
| | b. Fold pocket under 5/16 to 3/8 inch and double stitch, 3/16 to 1/4 inch gauge through all plies of material, the front and bottom edges of the pocket pieces. | 301 or 401 | LSd-2 | 10-14 | 30<br>30 | 30<br>50 |
| | c. Backstitch both ends of the pocket opening. Stitch across the facing width, commencing from the inner row of stitching approximately 3 inches below the waist raw edge, diagonally upward to outer top stitching (see operation 4.c), continuing the stitching as edgestitching superimposed on top stitching, to the waist raw edge, a distance of approximately 2-3/8 inches. Measure pocket opening and stitch superimposed on top stitching from bottom of pocket to pocket opening and diagonally across facing to inner row of facing stitching. | 301 or 401 | SSv-1 | 10-14 | 30<br>30 | 30<br>50 |
| 6. | Make hip patch pocket flap and hem. | | | | | |
| | a. Finished appearance. The flap and hem of the pocket shall be uniform in shape and size and shall finish without any twist, pleat or pucker. | | | | | |
| | b. Fold flap in half lengthwise as indicated by marks on pattern and stitch the side edges, 3/8 inch from edge. Trim corners, turn to finished position, work out points and topstitch the turned and folded edges, 1/8 inch from edge. | 301 | SSe-2 and OSf-1 | 10-14 | 30 | 30 |



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TABLE I - CONSTRUCTION OF TROUSERS

| No. | Operation | Stch Type | Seam/ Stch type | Stch in | Thread Ndl | Thread Bob/ Lpr |
|---|---|---|---|---|---|---|
| | OR | | | | | |
| | When alternate two piece pocket flap construction is used, turn in side and bottom edges of flap pieces and join plies with a double row of stitching. Stitching shall be 3/16 to 1/4 inch gauge with outer row of stitching 1/16 inch from turned edge. Edges of pocket flap plies may be precreased. | 301 | SSc-2 | 10-14 | 30 | 30 |
| | c. Stitch button hole (see operation 17.b). | | | | | |
| | d. Form hem of pocket as indicated by marks on pattern. Fold under raw edge 3/8 inch and stitch 1/16 to 1/8 inch from the turned edge. | 301 or 401 | EFb-1 | 10-14 | 30<br>30 | 30<br>50 |
| 7. | Join back darts and left hip pocket. | | | | | |
| | a. Finished appearance. The darts shall be carefully tapered to avoid unsightly bulging. Pocket may be precreased prior to setting. | | | | | |
| | b. Stitch the cut out dart in each back section, from 3/8 inch at the waist edge tapering to point at the bottom end. The thread ends shall be trimmed to 1/2 inch in length. When an automatic dart machine is used, the bottom end of the dart shall be backstitched with a minimum of 4 stitches. Turn and raise stitch, 1/16 inch from folded edge, with the seam allowance turned towards the center back. | 301 | LSq-2 | 10-14 | 30 | 30 |
| | c. Set and stitch, 1/8 inch from edge, the raw edge of the pocket flap over dart to the left back section as indicated by marks on pattern with the folded edge facing the waist. | 301 or 401 | LSb1-2(a) | 10-14 | 30<br>30 | 30<br>50 |
| | d. Fold under 5/16 to 3/8 inch and double stitch 3/16 to 1/4 inch gauge through all plies of material, the side and bottom edges of the pocket piece with the fold at the top of the hem 1/2 (± 1/8) inch below the flap raw edge. | 301 | LSd-2 | 10-14 | 30 | 30 |



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TABLE I - CONSTRUCTION OF TROUSERS

| No. | Operation | Stch Type | Seam/ Stch type | Stch in | Thread Ndl | Thread Bob/ Lpr |
|---|---|---|---|---|---|---|
| | e. The finished pocket opening shall measure 6-3/4 (± 1/4) inches wide for Type I; 6-1/4 (± 1/4) inches wide for Type II. | | | | | |
| | f. Turn flap to finished position and topstitch 3/16 to 1/4 inch from folded edge, enclosing the raw edges. | 301 | LSb1-2(b) | 10-14 | 30 | 30 |
| | g. Thr finished flap dimensions shall be 7 (± 1/4) inches wide and 2-1/2 (± 1/4) inches deep for type I; 6-1/2 (± 1/4) inches wide and 2-1/2 (± 1/4) inches deep for the type II. | | | | | |
| 8. | Type I only - stitch left and right flies and front crotch. | | | | | |
| | a. Finished appearance. The fly shall be of even length at waist and crotch seam without any twist, bulge or pucker when slide fastener is closed. The top edge of the slide fastener tape shall be diagonally turned under 2 (± 1/8) inches below the bound waist edge with the chain 1/2 to 3/4 inch in from the edge at top tapering to 1/4 to 3/8 inch in from the edge at bottom, 1/4 to 1/2 inch above the fly notch. | | | | | |
| | b. Stitch the front raw edge of the fly facing (left side) and the face side of the left lining fly edge, 1/8 to 3/16 inch from edge, the length of the fly. | 301 or 401 | SSa-1 | 10-14 | 30<br>30 | 30<br>50 |
| | c. Fold under 3/8 inch and topstitch the raw edge of the fly facing (left side), through the lining. | 301 or 401 | LSd-1 | 10-14 | 30<br>30 | 30<br>50 |
| | d. Double stitch, 5/16 to 3/8 inch gauge, the left half of the slide fastener tape to the left fly facing and lining, placing the bottom stop of the slide fastener about 3/8 inch above the fly notch. | 301 | SSv-2 | 10-14 | 30 | 30 |
| | e. Stitch face to face, 3/16 to 1/4 inch from edge, the lining left front to the basic material left front at fly edge, the length indicated by notch on pattern. Notch to stitch line. Turn to finished position and edgestitch 3/16 to 1/4 inch from folded edge. | 301 | SSe-2 | 10-14 | 30 | 30 |



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TABLE I - CONSTRUCTION OF TROUSERS

| No. | Operation | Stch Type | Seam/ Stch type | Stch in | Thread Ndl | Thread Bob/ Lpr |
|---|---|---|---|---|---|---|
| | f. Stitch, 3/8 inch from folded edge, the back turned edge of fly facing (left side) gradually curving to fly notch. The stitching shall be 1 5/8 (+ 1/8) inches from front edge of fly. | 301 | SSv-1 | 10-14 | 30 | 30 |
| | g. Stitch, 1/4 inch from edge, face side in, the fly extension (right side) pieces at the bottom edge. Turn, face side out, and continue the stitching along the raw edges of three remaining sides. | 301<br>401 | SSa-1 | 10-14 | 30<br>30 | 30<br>50 |
| | OR | | | | | |
| | As an alternate, front and bottom edges shall be stitched 1/4 inch from edge, turned and edgestitched 3/16 to 1/4 inch from folded edge. | 301 or<br>401 | SSe-2 | 10-14 | 30<br>30 | 30<br>50 |
| | h. Stitch buttonhole (see operation 17.c). | | | | | |
| | i. Stitch, 3/8 inch from edge, the right half of the slide fastener tape to the fly edge of the right front with top edge of the tape turned under and positioned to correspond with left half of slide fastener tape. | 301 | SSa-1 | 10-14 | 30 | 30 |
| | j. Join, 3/8 inch from edge, the left and right fronts at crotch. Turn seam to left and raisestitch with a double row of topstitching. | 301 or<br>401 | LSq-3 | 10-14 | 30<br>30 | 30<br>50 |
| | k. Join, 3/8 inch from edge, the assembled fly extension (right side) to the fly edge of the right front, the length indicated by the notch on pattern. | 301 or<br>401 | SSa-1 | 10-14 | 30<br>30 | 30<br>50 |
| | l. Join, 1/2 inch from edge, face to face, the lining right front to the right front at the fly edge from top of waist to fly notch, catching the prestitched assembled fly extension (right side). Slit notch at bottom of fly. Continue stitching, face to face, the lining fronts at crotch. | 301 or<br>401 | SSa-1 | 10-14 | 30<br>30 | 30<br>50 |



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TABLE I - CONSTRUCTION OF TROUSERS

| No. | Operation | Stch Type | Seam/ Stch type | Stch in | Thread Ndl | Thread Bob/ Lpr |
|---|---|---|---|---|---|---|
| | m. Turn back right front and lining right front, laying fly extension piece flat and smooth and raisestitch 1/16 inch from folded edge. | 301 | SSq-2(b) | 10-14 | 30 | 30 |
| | n. Tack the bottom edge of the fly extension (right side) to the face side of the lining left front with the lining crotch seam spread open and flat on the underside. | 301 | LSa-1 | 10-14 | 30 | 30 |
| 9. | Type II only - stitch left and right flies and front crotch. | | | | | |
| | a. Finished appearance. See operation 8a. | | | | | |
| | b. Stitch the front raw edge of the fly facing (right side) and the face side of the right lining fly edge, 1/8 to 3/16 inch from edge, the length of the fly. | 301<br>401 | SSa-1 | 10-14 | 30<br>30 | 30<br>50 |
| | c. Fold under 3/8 inch and topstitch the raw edge of the fly facing (right side), through the lining. | 301 or<br>401 | LSd-1 | 10-14 | 30<br>30 | 30<br>50 |
| | d. Double stitch, 5/16 to 3/8 inch gauge, the right half of the slide fastener tape to the right fly facing and lining, placing the bottom stop of the slide fastener about 3/8 inch above the fly notch. | 301 | SSv-2 | 10-14 | 30 | 30 |
| | e. Stitch, face to face, 3/16 to 1/4 inch from edge, the right lining front to the right front at fly edge, the length indicated by notch on pattern. Notch to stitch line. Turn to finished position and edgestitch 3/16 to 1/4 inch from folded edge. | 301 | SSe-2 | 10-14 | 30 | 30 |
| | f. Stitch 3/8 inch from folded edge the back turned edge of the fly facing (right side), gradually curving to fly notch. The stitching shall be 1-5/8 ($\pm$ 1/8 ) inches from front edge of fly. | 301 | SSv-1 | 10-14 | 30 | 30 |



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TABLE I - CONSTRUCTION OF TROUSERS

| No. | Operation | Stch Type | Seam/ Stch type | Stch in | Thread Ndl | Thread Bob/ Lpr |
|---|---|---|---|---|---|---|
| | g. Stitch, 1/4 inch from edge, face side in, the fly extension (left side) pieces at the bottom edge. Turn, face side out, and continue the stitching along the raw edges of the three remaining sides. | 301 or 401 | SSa-1 | 10-14 | 30<br>30 | 30<br>50 |
| | OR | | | | | |
| | As an alternate, front and bottom edges shall be stitched 1/4 inch from edge, turned and edgestitched 3/16 to 1/4 inch from folded edge. | 301 or 401 | SSe-2 | 10-14 | 30<br>30 | 30<br>50 |
| | h. Stitch buttonhole (see operation 17.c). | | | | | |
| | i. Stitch, 3/8 inch from edge, the left half of the slide fastener tape to the fly edge of the left front with top edge of the tape turned under and positioned to correspond with right half of slide fastener tape. | 301 | SSa-1 | 10-14 | 30 | 30 |
| | j. Join, 3/8 inch from edge, the right and left fronts at crotch. Turn seam to the right and raisestitch with a double row of topstitching. | 301 or 401 | LSq-3 | 10-14 | 30<br>30 | 30<br>50 |
| | k. Join, 3/8 inch from edge, the assembled fly extension (left side) to the fly edge of the left front, the length indicated by the notch on pattern. | 301 or 401 | SSa-1 | 10-14 | 30<br>30 | 30<br>50 |
| | l. Join, 1/2 inch from edge, face to face, the left lining front to the left front at the fly edge from top of waist to fly notch, catching the prestitched assembled fly extension (left side). Slit notch at bottom of fly. Continue stitching, face to face, the lining fronts at crotch. | 301 or 401 | SSa-1 | 10-14 | 30<br>30 | 30<br>50 |



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TABLE I - CONSTRUCTION OF TROUSERS

| No. | Operation | Stch Type | Seam/ Stch type | Stch in | Thread Ndl | Thread Bob/ Lpr |
|---|---|---|---|---|---|---|
| | m. Turn back left front and left lining front, laying fly extension piece flat and smooth and raisestitch 1/16 inch from folded edge. | 301 | SSq-2(b) | 10-14 | 30 | 30 |
| | n. Tack the bottom edge of the fly extension (left side) to the face side of the right lining front with the lining crotch seam spread open and flat on the underside. | 301 | LSa-1 | 10-14 | 30 | 30 |
| 10. | Join seat seam, outseam, and inseams. | | | | | |
| | a. Finished appearance. Seams joining the seat seam outseams and inseams shall be uniformly double stitched and carefully assembled to avoid any twist, pleat or pucker, and finished evenly. The inseams shall be joined in one continuous operation with the crotch and seat seam junction staggered not more than 3/8 inch from center of seam to center of seam. The stay (knee) piece may be stitched before the outseam and inseam are joined. | | | | | |
| | b. The direction of joining seams shall be as follows:<br>- Fronts over back at outseams and inseams.<br>- Left back over right back at seat seams.<br>- Center fronts lapped in accordance with fly. | | | | | |
| | c. Join seat seam at center back with a double lapped and double stitched seam. | 301 or 401 | LSc-2 | 10-14 | 30<br>30 | 30<br>50 |
| | d. Stitch, 1/8 to 3/16 inch from edge, the top edge of the back waist interlining to the waist edge, centered horizontally across the seat seam. | 301 or 401 | SSa-1 | 10-14 | 30<br>30 | 30<br>50 |
| | e. Join outseams, matching notches, with a double lapped and double stitched seam simultaneously inserting and catching one end of a base fabric stay piece at the knee. | 301 or 401 | LSc-2 | 10-14 | 30<br>30 | 30<br>50 |



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| No. | Operation | Stch Type | Seam/ Stch type | Stch in | Thread Ndl | Thread Bob/ Lpr |
|---|---|---|---|---|---|---|
| | f. Join inseams, matching notches, with a double lapped and double stitched seam simultaneously inserting and catching one end of a base fabric stay piece at the knee. | 301 or 401 | LSc-2 | 10-14 | 30<br>30 | 30<br>50 |
| | OR | | | | | |
| | As an alternate, operation 10.c, 10.e, and 10.f may be performed by using seam type LSq-3. | | | | | |
| 11. | Join pile darts, seat seam, outseams inseams, bottom leg lining extension and labels. | | | | | |
| | a. Stitch the cut out dart in each back section face to face, from 3/8 inch at the waist edge tapering to a point at the bottom end. | 301 | SSa-1 | 10-14 | 30 | 30 |
| | b. Stitch, 1/2 inch from edge, the seat seam, face to face, at center back. Press seam open and flat. | 301 or 401 | SSa-1 | 10-14 | 30<br>30 | 30<br>50 |
| | c. Stitch the four sides of the identification-instruction label centered horizontally across the seat seam with the top edge of the label approximately 2 inches below the turned edge of the binding on the finished trousers. The ends of the label stitching shall be over-lapped with not less than 3 stitches. | 301 | LSbj-1 | 10-14 | 30 | 30 |
| | d. Position the size/stock number label directly below the combination label and stitch around the four sides. | 301 | LSbj-1 | 10-14 | 30 | 30 |



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TABLE I - CONSTRUCTION OF TROUSERS

| No. | Operation | Stch Type | Seam/ Stch type | Stch in | Thread Ndl | Thread Bob/ Lpr |
|---|---|---|---|---|---|---|
| | NOTE: (1) As an alternate, stitch labels with box tacker. | 301 | LSbj-1 | 8-12 | 30 | 30 |
| | (2) The stay (knee) piece may be stitched after performing operations 11.e and 11.g. | | | | | |
| | e. Join, face to face, 1/2 inch from edge, matching notches, the lining outseams simultaneously superimposing and catching the free end of the base fabric stay piece at the knee. Press seam open and flat. | 301 or 401 | SSa-1 | 10-14 | 30<br>30 | 30<br>50 |
| | f. Stitch, 3/8 inch from edge, face to face, the top edge of the bottom leg lining extension piece to the lining side of the front and back sections. Turn and raisestitch, with the seam allowance on the underside of the bottom leg lining extension piece. | 301 or 401 | LSq-2 | 10-14 | 30<br>30 | 30<br>50 |
| | g. Join, face to face, 1/2 inch from edge, matching notches, the lining inseams simultaneously superimposing and catching the free end of the base fabric stay piece at the knee and the ends of the bottom leg lining extension piece. Press seam open and flat. | 301 or 401 | SSa-1 | 10-14 | 30<br>30 | 30<br>50 |
| 12. | Complete trouser assembly. | | | | | |
| | a. Turn trousers to finished position. Stitch 1/8 to 3/16 inch from edge, the two assemblies together at the waist edge simultaneously matching seams and back darts and catching the ends of the suspender (braid) loops on the basic material side of each back section, forming an opening of 2 1/4 (+ 1/4) inches with one end of the loop adjacent to the outseam and the other adjacent to the back dart seam. | 301 or 401, 504, 505 | SSa-1 | 10-14<br><br>8-10 | 30<br>30<br>70<br>70 | 30<br>50<br>70<br>70 |



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TABLE I - CONSTRUCTION OF TROUSERS

| No. | Operation | Stch Type | Seam/ Stch type | Stch in | Thread Ndl | Thread Bob/ Lpr |
|---|---|---|---|---|---|---|
| | b. Stitch, 1/8 to 3/16 inch from edge, the two assemblies together at the leg opening edge. | 301 or 401, 504, or 505 | SSa-1 | 10-14<br><br>8-10 | 30<br>30<br>70<br>70 | 30<br>50<br>70<br>70 |
| | c. Stitch (quilt) riser at waist edge between the back darts with five horizontal rows of stitching through all plies of material. The top row of quilt stitching shall be spaced 5/16 (+ 1/16) inch below the top raw edge and remaining four rows shall be evenly spaced 1/4 to 5/16 inch apart. | 301 | SSv-5 | 10-14 | 30 | 30 |
| | d. Stitch, 1/16 inch from folded edge, the top edge of the bottom leg lining extension piece through all plies of material. | 301 | SSv-1 | 10-14 | 30 | 30 |
| 13. | Make and attach leg adjustment (take up) strap assemblies.<br>a. Finished appearance. The leg adjustment (take-up) strap assemblies shall be horizontally aligned on the outseam. | | | | | |
| | b. Thread, fold, position, and stitch free end of long strap assemblies in accordance with figures 1 and 2. | Box-stch | EFb-1 | 10-14 | 30 | 30 |
| | c. Turn under both ends of the short strap with the folded edge located on each front section in accordance with pattern template. Stitch through all plies in accordance with figures 1 & 2. | X-box-stch | LSd-1 | 10-14 | 30 | 30 |
| | d. Turn under unhemmed end of the long strap with the folded edge located on each back section in accordance with pattern template. Stitch through all plies in accordance with figures 1 & 2. | X-box-stch | LSd-1 | 10-14 | 30 | 30 |



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TABLE I - CONSTRUCTION OF TROUSERS

| No. | Operation | Stch Type | Seam/ Stch type | Stch in | Thread Ndl | Thread Bob/ Lpr |
|---|---|---|---|---|---|---|
| | e. The finished length of the leg adjustment strap assembly for all sizes shall be as follows:<br><br>Short strap - 2 (+ 1/4) inches<br>Long strap - 8 3/4 (+ 1/4) inches<br><br>Short strap shall be measured from folded edge to fold over buckle bar.<br>Long strap shall be measured from outer folded edge of hem to turn under. | | | | | |
| 14. | Binding.<br>a. Finished appearance. The bindings shall measure 3/8 (+ 1/16) inch wide with the raw edges securely caught in the stitching. Not more than two splices shall be permitted. | | | | | |
| | b. Bind leg opening leaving an extension of approximately one inch at each inseam. | 301 | BSc-1 | 10-14 | 30 | 30 |
| | c. Type I only - Bind waist and right front fly in one continuous operation leaving an extension of approximately one inch at the left front fly edge and bottom front edge of fly extension (right side).<br><br>OR<br><br>When alternate operation is used (see operation 8.g), the binding shall terminate at the top front edge of the fly extension (right side) leaving an extension of approximately 1 inch. | 301 | BSc-1 | 10-14 | 30 | 30 |
| | d. Type II only - Bind waist and left front fly in one continuous operation leaving an extension of approximately one inch at the right front fly edge and bottom front edge of fly extension (left side).<br><br>OR<br><br>When alternate operation is used (see operation 9.g), the binding shall terminate at the top front edge of the fly extension (left side) leaving an extension of approximately 1 inch. | 301 | BSc-1 | 10-14 | 30 | 30 |



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TABLE I - CONSTRUCTION OF TROUSERS

| No. | Operation | Stch Type | Seam/ Stch type | Stch in | Thread Ndl | Thread Bob/ Lpr |
|---|---|---|---|---|---|---|
| 15. | Make and attach belt loops. | | | | | |
| | a. Finished appearance. The six belt loops shall finish 5/8 to 3/4 inch wide and shall be located in accordance with pattern marker. | | | | | |
| | b. Fold stripping (loop) in half lengthwise, face side out, turn under the raw edges parallel to the fold and stitch 1/16 to 1/8 inch from folded edge. | 301 or 401 | EFp-2 | 10-14 | 30<br>30 | 30<br>50 |
| | OR | | | | | |
| | c. As an alternate, stitch 1/16 to 1/8 inch from folded edge, face side out, the two plies of stripping (loop) with the side raw edges turned in. | 301 or 401 | EFn-2 | 10-14 | 30<br>30 | 30<br>50 |
| | d. Turn under and tack 1/8 inch from folded edge across the width, the belt loops at the top and bottom ends through all plies of material. The distance between the tacking shall be not less than 2 inches nor more than 2 1/4 inches. | 301 | LSd-1 | 10-14 | 30 | 30 |
| | OR | | | | | |
| | e. As an alternate, stitch a 3/4 inch (or two 3/8 inch) horizontal bartack as indicated above (see operation 16.b). | | | | | |
| 16. | Bartacking. | | | | | |
| | a. Finished appearance. The bartacks shall be well made and correctly placed when used as the only means of attaching an assembly or to reinforce points of strain as specified. The bartack at the bottom end of the front patch pocket opening shall extend beyond the inner row of the double stitched outseam. | | | | | |
| | b. Stitch a 3/8 to 1/2 inch bartack at the following locations: | Bartack | | 28 per bartack | 30 | 30 |



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TABLE I - CONSTRUCTION OF TROUSERS

| No. | Operation | Stch Type | Seam/ Stch type | Stch in | Thread Ndl | Thread Bob/ Lpr |
|---|---|---|---|---|---|---|
| | Type I only<br>- One, at the base of fly at junction of the crotch and left fly stitching. The bartack may be diagonal in line with left fly stitching or may be horizontal and shall extend across the crotch seam. | | | | | |
| | - One, across the bound edge of left front waist (left front fly) and bottom front edge of fly extension (right side) with the extended end of the binding turned to the underside and caught in the bartack. | | | | | |
| | Type II only<br>- One, at the base of fly at junction of the crotch and right fly stitching. The bartack may be diagonal in line with right fly stitching or may be horizontal and shall extend across the crotch seam. | | | | | |
| | - One, across the bound edge of the right front waist (right front fly) and bottom front edge of fly extension (left side) with the extended end of the binding turned to the underside and caught in the bartack. | | | | | |
| | Type I and II<br>- One, perpendicular to the top and bottom ends of the front patch pocket opening (see operation 4.a and 5.c). | | | | | |
| | - One, (3/4 inch) or two (3/8 inch) across the width at top and bottom folded edges of belt loop (see operation 15.e). | | | | | |
| | - One, across the bound edge of leg opening with the extended end of the binding turned to the underside and caught in the bartack. | | | | | |



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TABLE I - CONSTRUCTION OF TROUSERS

| No. | Operation | Stch Type | Seam/ Stch type | Stch in | Thread Ndl | Thread Bob/ Lpr |
|---|---|---|---|---|---|---|
| 17. | Make buttonholes. | | | | | |
| | a. Finished appearance. Buttonholes shall be 7/8 to 1 inch cut, eyelet end taper bar type worked over gimp with the ends of the buttonhole stitching and gimp securely tacked. Buttonholes shall be clean cut, well made and correctly positioned. The stitching shall be securely caught in the fabric with the purling on the outside. | Btnhl | | 52-56 per btnhl including tack | 30 | 50 |
| | b. Stitch a vertical buttonhole in center of hip pocket flap (1/4 inch off tolerance allowed) with the inside purled edge 1/2 (+ 1/8) inch in from the folded edge. | | | | | |
| | Type I only<br>c. Stitch a diagonal buttonhole in right front with the inside purled edge 3/8 (+ 1/8) inch in from the turned edge of binding at the front and waist edge of fly extension (right side). | | | | | |
| | d. Stitch a horizontal buttonhole in l e f t front with inside purled edge 1/2 (+ 1/8) inch in from fly edge and 3/4 (+ 1/8) inch below the turned under edge of the binding at waist edge. | | | | | |
| | Type II only<br>e. Stitch a diagonal buttonhole in left front with the inside purled edge 3/8 (+ 1/8) inch in from the turned edge of binding at the front and waist edge of fly extension (left side). | | | | | |
| | f. Stitch a horizontal buttonhole in right front with inside purled edge 1/2 (+ 1/8) inch in from fly edge and 3/4 (+ 1/8) inch below the turned under edge of the binding at waist edge. | | | | | |



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TABLE I - CONSTRUCTION OF TROUSERS

| No. | Operation | Stch Type | Seam/ Stch type | Stch in | Thread Ndl | Thread Bob/ Lpr |
|---|---|---|---|---|---|---|
| 18. | Mark and sew buttons. | | | | | |
| | a. Finished appearance. The button shall be properly aligned with buttonholes. Hand or machine sewn buttons shall have the thread ends hidden and securely tacked off with no less than two stitches. | Hand | | 4 per button double thread | 24 | |
| | | or | | | | |
| | | Machine | | 14-16 per button or | 24 | 24 |
| | | | | 20-24 per button | 24 | |
| | b. Sew a 30 line button through the hip pocket hem to correspond with the buttonhole in flap. | | | | | |
| | Type I only<br>c. Sew a 30 line button through fly extension (right side) to correspond with buttonhole in left front fly. | | | | | |
| | d. Sew a 30 line button on the inside of left front, through all plies, to correspond with buttonhole in fly extension (right side). | | | | | |
| | Type II only<br>e. Sew a 30 line button through the fly extension (left side) to correspond with buttonhole in right front fly. | | | | | |
| | f. Sew a 30 line button on the inside of right front, through all plies, to correspond with buttonhole in fly extension (left side). | | | | | |
| 19. | Cleaning.<br>Trim all ends of stitching and remove all loose threads. Trim ends of belt loops, remove all spots and stains and all shade and size markings. | | | | | |
| NOTE: | Caution shall be made so not to harm the material when trimming. | | | | | |
| 20. | Close trousers.<br>Close front fly slide fastener and completely button waist and left hip pocket. | | | | | |



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3.8 Sizes and measurements. The sizes and measurements of the finished trousers shall be in accordance with table II and table III. The measurements and tolerances are in inches. The measurements shall be taken as indicated below (see figure 1).

Table II - Sizes and measurements of trousers, Type I

| Size | | Waist (A) | Seat (B) | Inseam (C) |
|---|---|---|---|---|
| Small | (27-30) | 31 1/2 | 44 | 29 |
| Medium | (31-34) | 35 1/2 | 47 | 29 1/2 |
| Large | (35-38) | 39 1/2 | 51 | 29 1/2 |
| X-Large | (39-42) | 43 1/2 | 55 | 29 1/2 |
| XX-Large | (43-46) | 47 1/2 | 59 | 29 1/2 |
| Tolerance | | -1,+1/2 | -1,+1/2 | ± 1/2 |

Table III - Sizes and measurements of trousers, Type II

| Size | | Waist (A) | Seat (B) | Inseam (C) |
|---|---|---|---|---|
| Small | (22-25) | 26 | 38 | 31 |
| Medium | (26-29) | 30 | 42 | 31 |
| Large | (30-33) | 34 | 46 | 31 |
| Tolerance | | -1,+1/2 | -1,+1/2 | ± 1/2 |

NOTE: Trousers shall be laid flat and smooth with the slide fastener, and buttons and buttonholes of front fly completely closed.

Waist (A) - Double the measurement taken straight across the front of trousers from folded edge to folded edge at a point on each side located 1 inch below the top of trousers at center front.

Seat (B) - Double the measurement taken straight across the front of trousers from folded edge to folded edge at a point 1 inch above the trouser fly bartack.

Inseam (C) - The measurement shall be taken along the inseam from the junction of the front center seam and the inseams to the bottom of the trousers.

3.9 Workmanship. The finished trousers shall conform to the quality of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.



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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the government to acceptance of defective material.

4.1.2 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

4.3 First article inspection. The first article submitted in accordance with 3.2 shall be inspected as specified in 4.4.2 for compliance with design, construction, workmanship and dimensional requirements.

4.4 Quality conformance inspection. Inspection shall be performed in accordance with MIL-STD-1488, except when otherwise indicated.

4.4.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase documents. In addition, the supplier shall furnish a certificate of compliance for the enamel finish for the buckle specified in 3.3.7. A certificate of compliance shall also be submitted stating that all components of the slide fastener are furnished from the same supplier (see 3.3.6).



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4.4.2 End item examination. The end item examination provisions for finished trousers shall be as specified for utility ("U") trousers in MIL-STD-1488 and with the following additional defects:

| Defect | Point Value |
|---|---|
| FLY EXTENSION ON LEFT SIDE (TYPE II ONLY) | |
| a. Bottom edge not caught in lining crotch seam | 3 |
| b. Tacking of bottom edge to lining left front: | |
| - omitted or insecure | 2 |
| - less than 1 1/2 inches from lining crotch seam | 1 |

4.4.3 Palletization examination. An examination shall be made to determine that the palletization complies with section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one palletized unit load fully packaged. The lot size shall be the number of palletized unit loads in the end item inspection lot. The inspection level shall be S-1 and the AQL, expressed in defects per hundred units, shall be 6.5.

| Examine | Defect |
|---|---|
| Finished dimensions | Length, width or height exceeds specified maximum requirement |
| Palletization | Pallet pattern not as specified.<br>Interlocking of loads not as specified.<br>Load not bonded with required straps as specified. |
| Weight | Exceeds maximum load limit. |
| Marking | Omitted, incorrect, illegible, of improper size, location, sequence, or method of application. |

5. PACKAGING

5.1 Preservation - packaging. Preservation-packaging shall be of level A or C (see 6.2).

5.1.1 Level A. Each pair of trousers shall be folded as follows:

1. Close front fly.
2. Fold trousers in half lengthwise with the fly inside the fold.
3. Fold trousers in half by bringing the bottom up to the top of the garment.

The approximate size of the folded garment shall be 14 1/2 inches by 23 inches.

5.1.2 Level C. Trousers shall be packaged to afford adequate protection against corrosion and physical damage during shipment from the supply source to the first receiving activity. The contractor may use his commercial practice when it meets these requirements.



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5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Ten (10) pairs of trousers of one size, class and type only, packaged as specified in 5.1, shall be packaged in a fiberboard shipping container, assembled, closed and reinforced conforming to type CF, class weather resistant, variety DW, grade V15c, size 3A of MIL-B-17757. The liner will be type CF, class domestic, in accordance with the requirements specified in MIL-B-17757. Level A packages shall be packed flat one in length, one in width, and ten in depth within a shipping container. Each container shall have the contents completely covered on the top and bottom with a sheet of commercial grade kraft paper. Toward the end of the contract or order, or when there are less than the required amount per container of the same size, mixed sizes may be packed within the same shipping container.

5.2.2 Level B. Ten (10) pairs of trousers of one type, class and size only, packaged as specified in 5.1, shall be packed in a fiberboard shipping container assembled and closed conforming to type CF, class domestic, variety DW, grade 275, size 3A of MIL-B-17757. The liner will be type CF, class dometic, in accordance with the requirements specified in MIL-B-17757. Level A packages shall be packed flat one in length, one in width and ten in depth within a shipping container. Each container shall have the contents completely covered on the top and bottom with a sheet of commercial grade kraft paper. Toward the end of the contract or order or when there are less than the required amount per container of the same size, mixed sizes may be packed within the same shipping container.

5.2.3 Level C. Trousers packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. The quantity per shipping container shall be the same as that normally used by the contractor for retail distribution. Containers shall be in accordance with Uniform Freight Classification Rules, National Motor Freight Classification Rules or US Postal Service Manual, as applicable.

5.3 Marking. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

5.3.1 Labels, mixed sizes. Each shipping container, packed with mixed sizes only, shall have securely attached to the end and side, directly under the printing or stenciling, a white paper label 5 by 4 inches with the words "MIXED NSN" plainly stamped or printed thereon, and under these words shall be legibly stampled or printed the correct quantity and National Stock Numbers contained therein.



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5.4 Palletization. When specified (see 6.2), item, packed as specified in 5.2, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Each prepared load shall be banded with primary and secondary straps in accordance with the banding means C, K and L, or O or P. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. The pallet shall be 4-way, Type IV; Type V, class 1, size 2; or Type VIII, fabricated from wood groups I, II, III or IV, Grade A of NN-P-71, or 4-way, Style 1, Size A, Type I, Class 1 fabricated from groups specified of MIL-P-15011. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contacting officer.

6. NOTES

6.1 Intended use. The trousers covered by this specification are intended to be worn in conjunction with the Jacket and Cap, Cold Weather, items as part of an outfit.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification
b. Type, class and size required (see 1.2).
c. When first article is required (see 3.2 and 4.3). The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, quantity and testing of the first article.
d. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
e. When palletization is required (see 5.4).

6.3 Samples and patterns. For access to samples and patterns, address the procuring activity issuing the invitation for bids.

6.4 Subject term (key word) listing.

Trousers, cold weather with lining
Trousers, fire retardant plain weave
Trousers, cotton/nylon sateen
Outfit, cold weather
Clothing, flame resistant
Clothing, wind resistant

Custodian:
Navy - NU

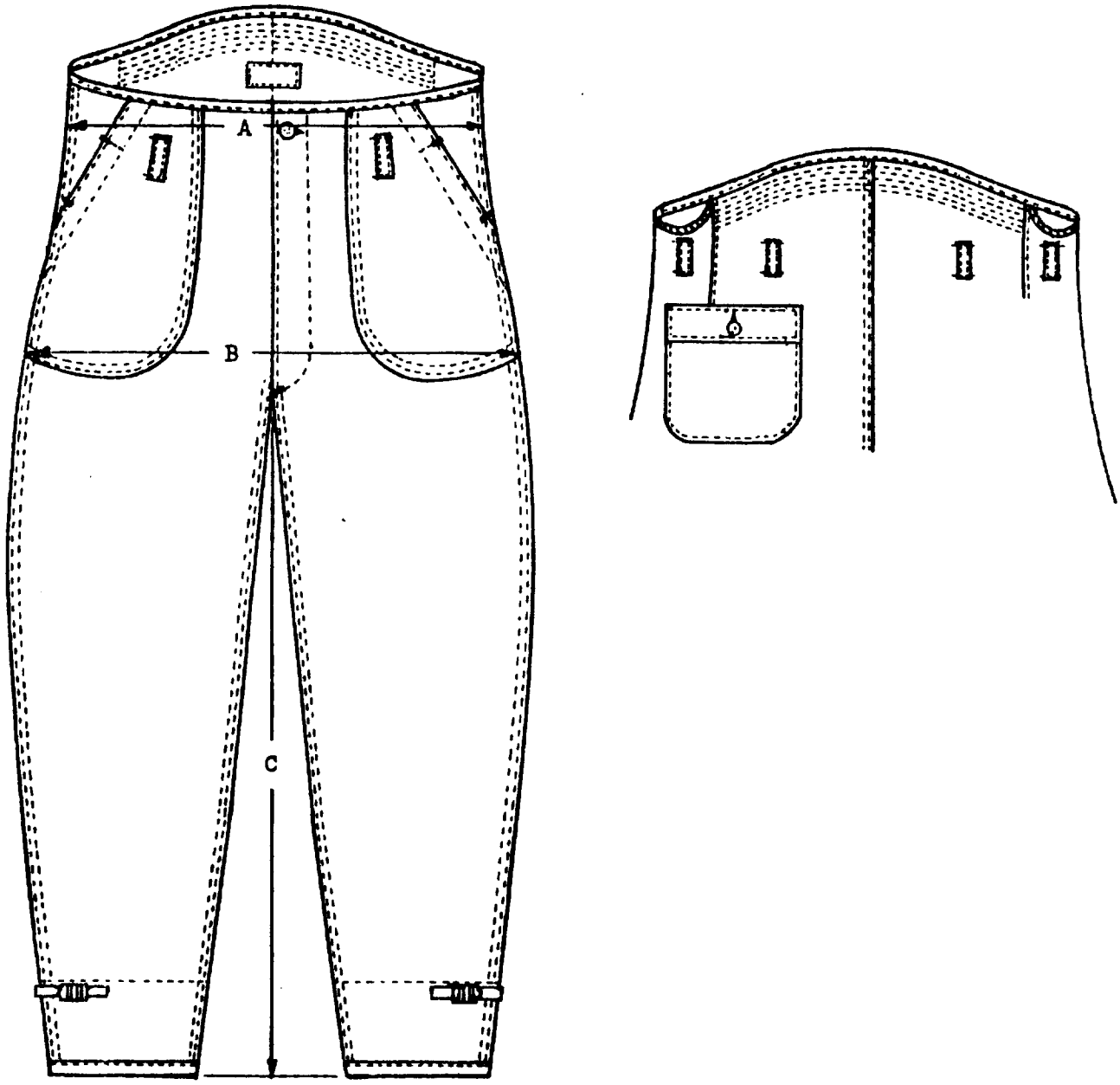
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Navy - NU

Review activity:
DLA-CT

Project no. 8415-0583



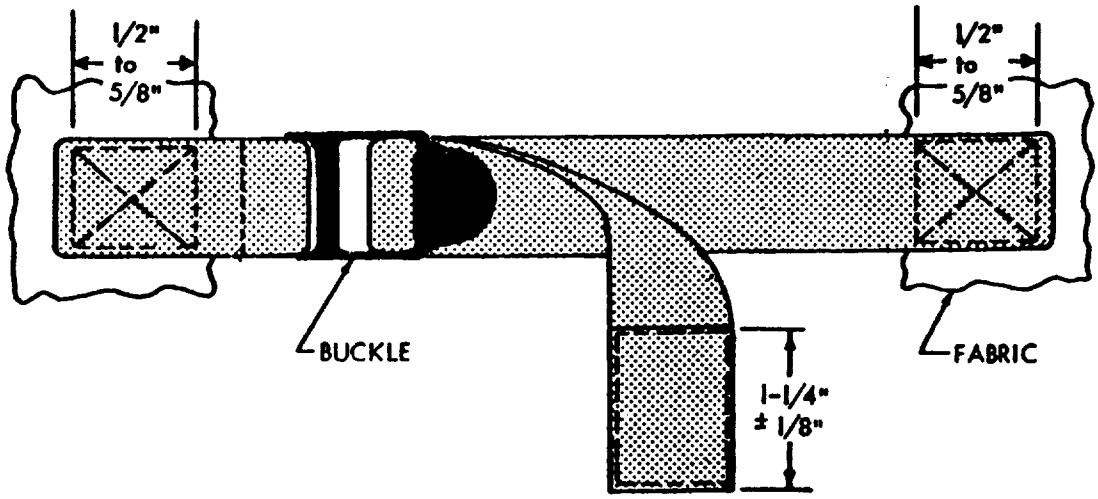
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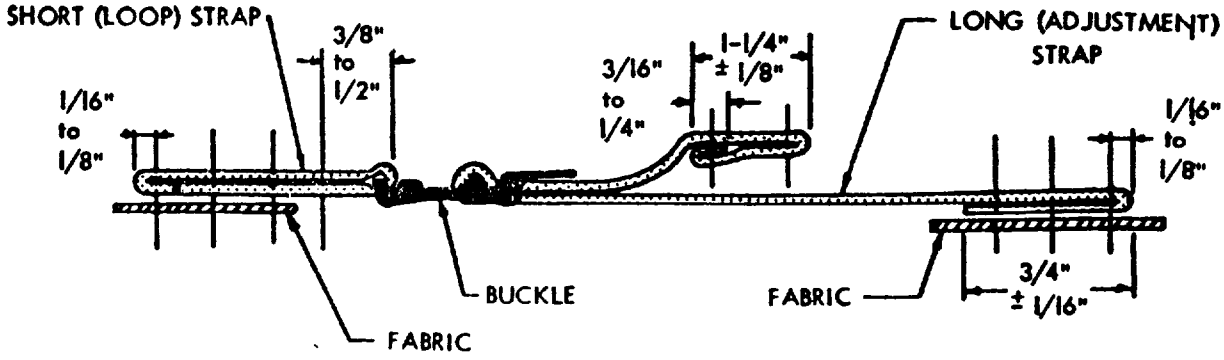

Type I

FIGURE 1 - TROUSERS, COLD WEATHER, FLAME RESISTANT

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**TOP VIEW**




**SIDE VIEW**

FIGURE 2 - LEG ADJUSTMENT STRAP AND BUCKLE ASSEMBLY



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| MIL-T-21704F | TROUSERS, COLD WEATHER |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR |
| | ☐ USER |
| b. ADDRESS *(Street, City, State, ZIP Code)* | ☐ MANUFACTURER |
| | ☐ OTHER *(Specify)*: _____ |

**5. PROBLEM AREAS**

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

**6. REMARKS**

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |

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